

Adapter Optimization Between Frozen Modules: A Controlled NTK Study

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Abstract

Modern speech-language systems often connect a frozen speech encoder to a frozen language model through a small trainable adapter. Empirically, the performance of such systems depends strongly on the particular encoder, adapter, and downstream model, but directly studying this interaction with full audio and LLM training is too expensive for a controlled theory project. This report isolates one component: the optimization of an adapter trained between two frozen modules. The full system is replaced by a teacher-student surrogate in which the encoder is a Gaussian representation distribution and the downstream model is a fixed head. The main claim is that adapter width alone does not determine optimization difficulty; the conditioning of the frozen representation and the frozen head enters through the empirical neural tangent kernel (NTK). In the surrogate, the NTK decomposes into terms involving the representation Gram matrix and the frozen-head Gram matrix. Experiments confirm a strong width effect and show that poor encoder or head conditioning decreases the relevant NTK eigenvalues and leaves a larger training error, especially in the moderate-to-wide adapter regime. Additional joint-conditioning and kernel-drift experiments show that the two frozen modules interact through the spectrum and that the empirical NTK becomes more stable as width increases, supporting the use of the initialization NTK as a diagnostic in the overparameterized regime.

1 Introduction

Pretrained speech-language models can be viewed as a composition

$$x \xrightarrow{E} z \in \mathbb{R}^{d_E} \xrightarrow{A_\theta} u \in \mathbb{R}^{d_L} \xrightarrow{\phi} y \in \mathbb{R}^{d_Y},$$

where E is a frozen audio encoder, ϕ is a frozen downstream model, and only the adapter A_θ is trained. This setting is motivated by empirical evidence that encoder rankings and adapter choices do not transfer cleanly across audio-language systems [Yang et al., 2021, Zhang et al., 2025, Lei et al., 2026]. This report focuses on the optimization aspect of this phenomenon: *when is adapter-only training easy or difficult, given fixed surrounding modules?*

The main simplification is to replace real audio and LLMs by a tractable surrogate that preserves the chain-rule structure of the end-to-end Jacobian. This follows a standard deep-learning-theory workflow: choose a phenomenon, strip it down to a model whose quantities are observable, and analyze that model. The resulting question is narrow but testable: how do the covariance of frozen representations and the Jacobian of the frozen head affect the empirical NTK and gradient-descent behavior of the adapter?

2 Problem Setup

The experimental dataset is $\mathcal{D} = \{(z_i, y_i)\}_{i=1}^n$, where z_i is the frozen encoder representation and y_i is a vector-valued label. The encoder is modeled by

$$z_i \sim \mathcal{N}(0, \Sigma_E).$$

Labels are generated by a teacher adapter $A_\star \in \mathbb{R}^{d_L \times d_E}$ and a fixed downstream head $\phi : \mathbb{R}^{d_L} \rightarrow \mathbb{R}^{d_Y}$:

$$y_i = \phi(A_\star z_i) + \xi_i, \quad \xi_i \sim \mathcal{N}(0, \sigma^2 I_{d_Y}).$$

The student adapter is a two-layer ReLU network

$$A_\theta(z) = \frac{1}{\sqrt{m}} W_2 \sigma(W_1 z), \quad W_1 \in \mathbb{R}^{m \times d_E}, \quad W_2 \in \mathbb{R}^{d_L \times m},$$

where m is the adapter width and $\theta = (W_1, W_2)$ are the only trainable parameters. The predictor is $f_\theta(z) = \phi(A_\theta(z))$, trained by full-batch gradient descent on

$$\hat{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \|f_\theta(z_i) - y_i\|_2^2.$$

Let $J_0 = \nabla_\theta f_{\theta_0}(\mathcal{D})$ be the Jacobian of the vectorized model outputs at initialization and $K_0 = J_0 J_0^\top$ the empirical NTK. The empirical representation covariance is $\hat{\Sigma}_z = n^{-1} \sum_i z_i z_i^\top$. The frozen-head sensitivity is summarized by

$$H_\phi = \frac{1}{n} \sum_{i=1}^n D\phi(u_i) D\phi(u_i)^\top, \quad u_i = A_{\theta_0}(z_i),$$

which becomes $H_\phi = CC^\top$ for a linear head $\phi(u) = Cu$. For a positive definite matrix M , $\kappa(M) = \lambda_{\max}(M)/\lambda_{\min}(M)$ denotes its condition number.

3 NTK Analysis

The optimization hypothesis is that adapter training is governed by the small eigenvalues of K_0 . The following standard NTK-style statement makes the role of this eigenvalue explicit [Jacot et al., 2018, Du et al., 2019].

Proposition 3.1 (Linearized NTK convergence). *Consider the linearized dynamics around initialization, where the residual vector evolves as $r_{t+1} = (I - \eta K_0)r_t$. If $\lambda_0 = \lambda_{\min}(K_0) > 0$ and $\eta \leq 1/\lambda_{\max}(K_0)$, then*

$$\|r_{t+1}\|_2^2 \leq (1 - \eta\lambda_0)^2 \|r_t\|_2^2.$$

Proof: Since K_0 is symmetric positive semidefinite, the eigenvalues of $I - \eta K_0$ lie in $[0, 1 - \eta\lambda_0]$ under the stated step-size condition. Therefore

$$\|r_{t+1}\|_2 = \|(I - \eta K_0)r_t\|_2 \leq (1 - \eta\lambda_0) \|r_t\|_2,$$

and squaring both sides gives the claim. $\square$

Standard wide-network NTK analyses justify using this linearized dynamics when the empirical kernel does not move much during training [Jacot et al., 2018, Du et al., 2019]. Thus the small eigenvalues of K_0 are a natural diagnostic for adapter optimization.

For the linear-head surrogate, the NTK can be written explicitly. Let $c_a^\top$ be row a of C , $w_{2,r}$ be column r of W_2 , $h_i = \sigma(W_1 z_i)$, and $s_{ir} = \mathbf{1}\{(W_1 z_i)_r > 0\}$. For samples i, j and output coordinates a, b ,

$$\begin{aligned} K_{0,(i,a),(j,b)} &= \frac{1}{m} \langle h_i, h_j \rangle (CC^\top)_{ab} \\ &\quad + \frac{1}{m} \sum_{r=1}^m s_{ir} s_{jr} \langle z_i, z_j \rangle (c_a^\top w_{2,r}) (c_b^\top w_{2,r}). \end{aligned}$$

This identity shows why both frozen modules matter. The representation Gram matrix enters through $\langle z_i, z_j \rangle$, while the downstream head enters through $CC^\top$ and the projected adapter columns $Cw_{2,r}$. Thus poor conditioning of either the representation distribution or the frozen head can shrink the useful spectrum of K_0 and require a larger width m to obtain comparable convergence. This analysis leads to three empirical predictions. First, increasing m should improve optimization by making the random adapter features more expressive and by stabilizing the empirical kernel. Second, worse conditioning of either $\hat{\Sigma}_z$ or H_ϕ should reduce small NTK eigenvalues and increase the residual error at fixed width. Third, if both frozen modules are ill-conditioned, the error should be worse than in the corresponding one-sided settings because both Gram factors enter the same Jacobian product. The experiments below test these predictions directly.

Experiment	Setting	$m = 16$	$m = 128$	$m = 512$
E1 train MSE	$\kappa_E = 1$	2.28×10^{-2}	1.59×10^{-3}	1.36×10^{-4}
E1 train MSE	$\kappa_E = 100$	1.52×10^{-2}	3.03×10^{-3}	5.93×10^{-4}
E2 train MSE	$\kappa_H = 1$	2.40×10^{-2}	1.56×10^{-3}	1.17×10^{-4}
E2 train MSE	$\kappa_H = 100$	1.96×10^{-2}	3.27×10^{-3}	1.78×10^{-3}

Table 1. Representative final training MSE values. Width improves optimization in both sweeps, but ill-conditioned frozen modules leave larger residual error at the same width.

4 Experiments

All experiments use generated representation-label data rather than raw audio. Unless stated otherwise, the default setting is $d_E = 64$, $d_L = 32$, $d_Y = 8$, $n_{\text{train}} = 512$, $n_{\text{val}} = 512$, $n_{\text{test}} = 4096$, and noise level $\sigma = 0.05$. Each configuration is averaged over five seeds. The reported optimization metric is final training MSE after 800 full-batch gradient steps with the learning rate capped at 1.0.

E1: encoder conditioning. The head is fixed with $\kappa(H_\phi) = 1$, while Σ_E is diagonal with log-spaced eigenvalues and $\kappa_E \in \{1, 3, 10, 30, 100\}$. The adapter width is $m \in \{16, 32, 64, 128, 256, 512\}$.

E2: frozen-head conditioning. The encoder distribution is isotropic, $\Sigma_E = I$, while the linear head C is generated so that $\kappa(CC^\top) \in \{1, 3, 10, 30, 100\}$. The same width grid is used.

E3: NTK predictiveness. E1 and E2 are pooled to test whether $\lambda_{\min}(K_0)$ predicts optimization error beyond width. The analysis computes correlations between $\log \lambda_{\min}(K_0)$ and $\log$ final training MSE and fits simple log-linear least-squares predictors.

E4: joint conditioning. The one-dimensional sweeps are not sufficient to test whether the two frozen modules interact. E4 therefore sweeps $(\kappa_E, \kappa_H) \in \{1, 10, 100\} \times \{1, 10, 100\}$ at widths $m \in \{128, 512\}$ using three seeds. This experiment checks whether the easiest and hardest regimes are organized by the joint spectrum of the representation and head factors rather than by either factor alone.

E5: kernel drift. The NTK analysis uses the initialization kernel K_0 , so E5 measures how far the empirical kernel moves during nonlinear training. For the well-conditioned setting $(\kappa_E, \kappa_H) = (1, 1)$, the ill-conditioned-encoder setting $(100, 1)$, and the ill-conditioned-head setting $(1, 100)$, the experiment computes

$$\frac{\|K_T - K_0\|_F}{\|K_0\|_F},$$

where K_T is the empirical NTK at the final iterate. Because this requires two exact NTK computations per run, E5 uses $n_{\text{train}} = 256$, $n_{\text{test}} = 2048$, $n_{\text{NTK}} = 32$, widths $m \in \{64, 128, 256, 512\}$, and three seeds.

5 Results

Table 1 and Figure 1 show the two main sweeps. Width is the dominant factor: increasing m from 16 to 512 reduces final training MSE by one to two orders of magnitude. Conditioning matters once the adapter is moderately wide. In E1, at $m = 512$, the final MSE increases from 1.36×10^{-4} at $\kappa_E = 1$ to 5.93×10^{-4} at $\kappa_E = 100$; the corresponding mean $\lambda_{\min}(K_0)$ decreases from 9.39 to 6.85. In E2 the effect is stronger: at $m = 512$, the final MSE increases from 1.17×10^{-4} at $\kappa_H = 1$ to 1.78×10^{-3} at $\kappa_H = 100$, while $\lambda_{\min}(K_0)$ decreases from 9.62 to 0.46.

Figure 2 shows that the NTK diagnostic is predictive, but not the only explanatory variable. Across all 300 E1/E2 runs, the Pearson correlation between $\log \lambda_{\min}(K_0)$ and $\log$ final train MSE is -0.759 . Width alone already explains much of the variance in this controlled experiment ($R^2 = 0.876$ for a log-linear width-only predictor), while adding conditioning information increases the fit to $R^2 = 0.916$. The NTK signal becomes strongest in the overparameterized regime: within the fixed width slice $m = 512$, the combined E1/E2 correlation between $\log \lambda_{\min}(K_0)$ and $\log$ final train MSE is -0.896 (and -0.969 in E2 alone). This is consistent with the theoretical picture: after width is large enough to remove severe underparameterization, the frozen-module spectrum becomes a meaningful predictor of residual optimization error.

The joint-conditioning sweep provides a stronger test of the hypothesis because it varies both frozen modules simultaneously. Figure 3 shows that increasing either κ_E or κ_H raises the final error and generally reduces $\lambda_{\min}(K_0)$, with the largest degradation when both are ill-conditioned. At $m = 512$, the well-conditioned setting has final train MSE 1.14×10^{-4} and mean $\lambda_{\min}(K_0) = 9.45$. With only encoder ill-conditioning $(\kappa_E, \kappa_H) = (100, 1)$, the MSE rises to 5.30×10^{-4} and $\lambda_{\min}(K_0)$ drops to 6.64. With only head ill-conditioning $(1, 100)$, the MSE rises further to 1.77×10^{-3} and $\lambda_{\min}(K_0)$ drops to 0.465.

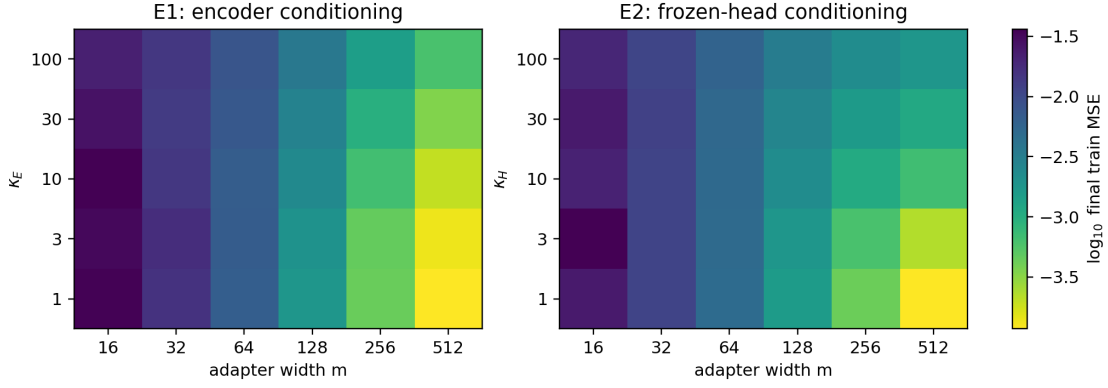


Figure 1. Final train MSE over width and conditioning. Darker values correspond to smaller MSE. Both encoder-side and head-side ill-conditioning make optimization harder at fixed width.

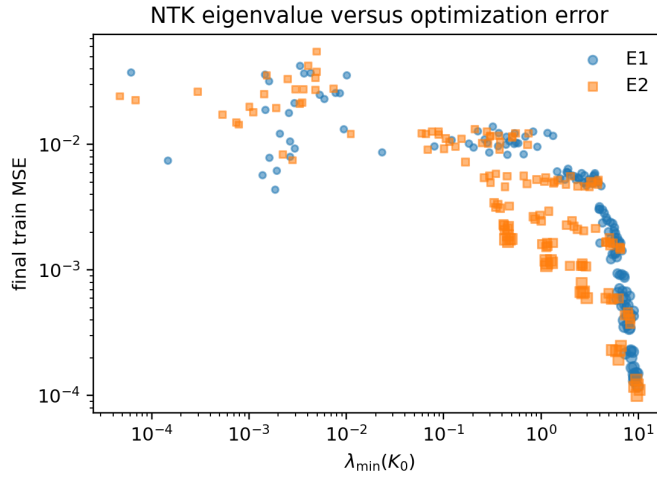


Figure 2. Relationship between the empirical NTK minimum eigenvalue and final training error for E1 and E2. Marker size increases with adapter width.

When both modules are ill-conditioned (100, 100), the MSE is 2.34×10^{-3} and $\lambda_{\min}(K_0)$ is only 0.306. Across the E4 runs, the correlation between $\log \lambda_{\min}(K_0)$ and $\log$ final train MSE is -0.793 at $m = 128$ and -0.915 at $m = 512$, again showing that the NTK spectrum becomes most informative in the wider regime.

E5 checks that the preceding NTK interpretation is most appropriate at larger width. As shown in Figure 4, the relative kernel drift decreases nearly monotonically with m in all three tested conditions. In the well-conditioned case, drift falls from 0.400 at $m = 64$ to 0.102 at $m = 512$; for the ill-conditioned encoder it falls from 0.446 to 0.122; for the ill-conditioned head it falls from 0.346 to 0.077. Relative parameter movement shows the same trend, decreasing from roughly 0.15–0.17 at $m = 64$ to about 0.05 at $m = 512$. Thus the wide adapters used to interpret $\lambda_{\min}(K_0)$ are also the settings where the nonlinear model stays closest to its initialization kernel. At the same time, the ill-conditioned-head setting still has much larger final training MSE at $m = 512$ (6.79×10^{-4} versus 6.12×10^{-5} in the well-conditioned E5 setting), indicating that the difficulty is not caused merely by kernel drift; it is already visible in the small useful spectrum of K_0 .

6 Discussion

The experiments support a specific interpretation of adapter optimization. At very small width, the student is simply underparameterized, so width dominates the error. Once width is moderately large, the adapter has enough random features to fit the teacher map, and the frozen modules become the main source of variation. This explains why $\lambda_{\min}(K_0)$ is most predictive within the wide slices of E3 and E4. In that regime, the model is not failing because it lacks parameters; it is failing because the gradient directions induced by the frozen encoder and head do not span the output residuals equally well.

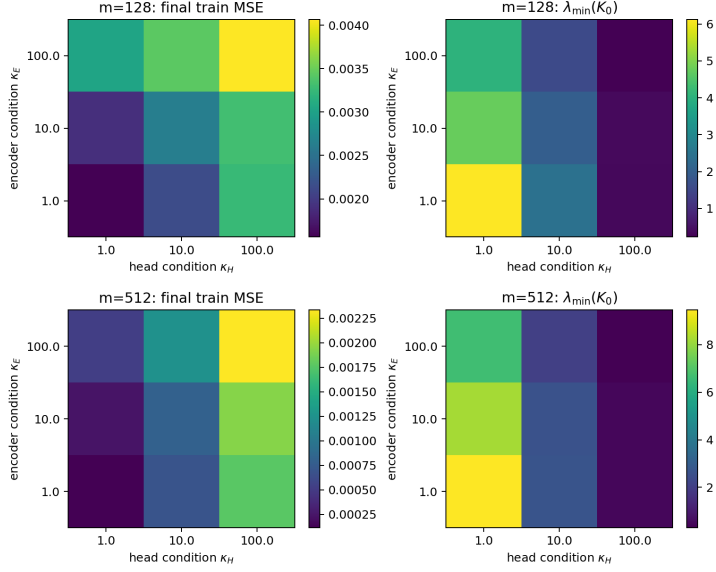


Figure 3. E4 joint-conditioning sweep. The left column reports final training MSE, and the right column reports $\lambda_{\min}(K_0)$. The hardest settings coincide with small useful NTK eigenvalues, especially at larger width.

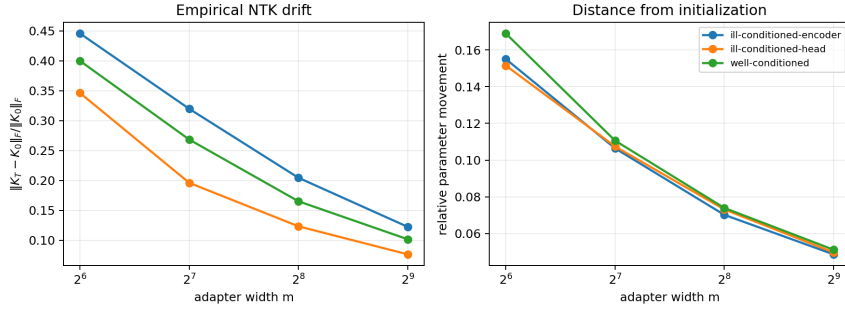


Figure 4. E5 kernel drift and parameter movement. Wider adapters move less in both parameter space and empirical-NTK space, making the initialization NTK a more reliable diagnostic.

7 Conclusion and Limitations

This project gives a controlled account of adapter optimization between frozen modules. The main result is qualitative but robust: adapter width is necessary, but the conditioning of the frozen representation and frozen head affects the empirical NTK spectrum and the remaining optimization error. The joint-conditioning experiment verifies that the two frozen modules are not independent nuisances; they combine through the same empirical kernel. The kernel-drift experiment further shows that this spectral diagnostic is most defensible in the moderate-to-wide regime, where the nonlinear adapter remains closer to its initialization kernel. Together, these results explain in a tractable model why different frozen encoder–LLM pairs may require different adapter widths even when the adapter architecture is unchanged.

The limitations are also clear. The experiments use Gaussian representations and fixed linear heads, not real audio encoders or LLMs. The binary success threshold 10^{-4} was too strict under the conservative learning-rate budget, so the final analysis relies on continuous training-error metrics. Future work should test the same NTK diagnostics on cached representations from real speech encoders and on nonlinear frozen heads with more realistic local Jacobian spectra.

References

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